AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are THIRTEEN questions. Answer TEN questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER Q.1 OR Q.2)

Q.1. (Total 5 Marks)

a. Decide whether $\sqrt{3} + x^2 + x^3$ is a polynomial or not. Give a reason for your answer. (2 Marks)

b. Find the value of $a - b$ by using appropriate formula if $a + b = 4$ and $ab = 4$. (3 Marks)

(ATTEMPT EITHER Q.1 OR Q.2)

Q.2. (Total 5 Marks)

a. Find $P(3)$ for $P(t) = t^3 - 5t - 17$. (2 Marks)

b. Find the continued product of $\left(x - \frac{1}{\sqrt{2}}\right)\left(x + \frac{1}{\sqrt{2}}\right)\left(x^2 + \frac{1}{2}\right)$. (2 Marks)

c. Rationalize the denominator of $\frac{2}{\sqrt{3}}$. (1 Mark)

(ATTEMPT EITHER Q.3 OR Q.4)

Q.3. (Total 5 Marks)

a. Verify that $x = 0$ and $x = \frac{-b}{a}$ are zeros of polynomial $P(x) = ax^2 + bx$. (3 Marks)

b. Factorize completely $27x^3 - 1$. (2 Marks)

(ATTEMPT EITHER Q.3 OR Q.4)

Q.4. (Total 5 Marks)

If $x-1$ is a factor of $P(x)=x^3-2x^2-5x+6$, then find other factors of $P(x)$.

Q.5. (Total 5 Marks)

a. Solve $\frac{3x-3}{5}=3x-3$. (2 Marks)

b. Solve and write the solution set of $3x-2\le 7x+12$, where $x\in N$. (3 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6)

Q.6. (Total 4 Marks)

a. If $x^4 - 2x^3 + 3x^2 - bx + 1$ is a perfect square, then find the value of *b*.

b. Simplify $\left[\frac{(x-y)}{x^3y - xy^3} \div \frac{1}{x+y}\right]\left(x^2y^2\right)$.

(ATTEMPT EITHER PART a OR PART b OF Q.7)

Q.7. (Total 5 Marks)

a. Solve $x^2 - \frac{2}{3}x = \frac{3}{2}x - 1$.

b. Prove by method of completing square that the roots of $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

(ATTEMPT EITHER Q.8 OR Q.9)

Q.8. (Total 5 Marks)

If $A=\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$, $B=\begin{bmatrix}2 & 4\\ 3 & 5\end{bmatrix}$ and $A\times B=\begin{bmatrix}11 & 19\\ 16 & 28\end{bmatrix}$, then prove that $(A\times B)^t = B^t \times A^t$.

(ATTEMPT EITHER Q.8 OR Q.9)

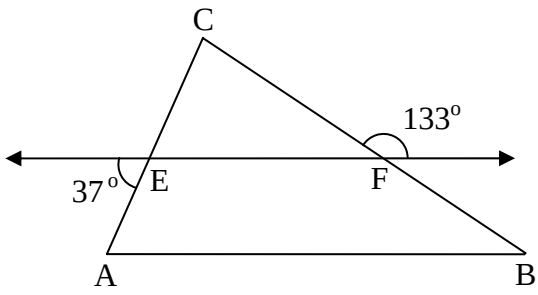
Q.9. (Total 5 Marks)

Column I contains names of the matrices. For each name in column I, find the appropriate matrices in column II. Record your answer in the space given before each name in column I.

Column I	Column II
_____1. Singular matrix	A. $\begin{bmatrix}1 & 3\\ 3 & 2\end{bmatrix}$
_____2. Symmetric matrix	B. $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$
_____3. Scalar matrix	C. $\begin{bmatrix}1 & 7\end{bmatrix}$
_____4. Column matrix	D. $\begin{bmatrix}1\\ 7\end{bmatrix}$
_____5. Identity matrix	E. $\begin{bmatrix}1 & 3\\ 2 & 6\end{bmatrix}$
	F. $\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$

Q.10. (Total 5 Marks)

In the figure given below, AB is parallel to EF. Find $m\angle A$, $m\angle B$ and $m\angle C$. Give reasons for your answers.



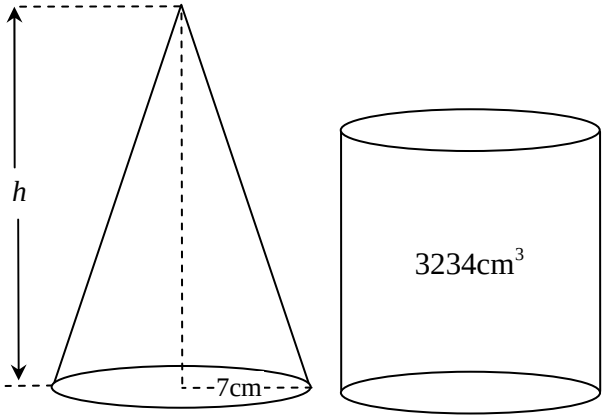
NOT TO SCALE

Q.11. (Total 4 Marks)

Draw a circle with centre O of radius 2.5 cm and take a point T at its circumference and draw a tangent to the circle passing through point T.

Q.12. (Total 4 Marks)

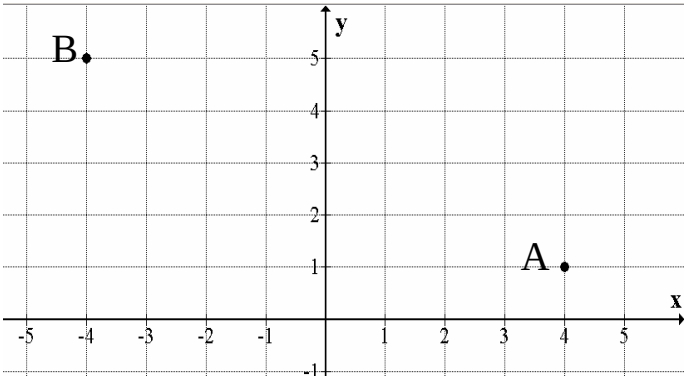
The radius of the base of the cone is 7cm and height is *h* cm. A cylinder has volume 3234cm^3. If the volume of the cone and the cylinder is equal, then find the value of *h*. $\left[\pi = \frac{22}{7}\right]$



NOT TO SCALE

Q.13. (Total 3 Marks)

For the following graph, find the distance between points A and B.



END OF PAPER

Please use this page for rough work